PBL – Week 7 (CIE)

Testing and Software Modules – Mediconnect Project

1. Understanding Testing Requirements

A. Patient Module

Functional Requirements

- Patient should be able to register and log in securely.
- Patient should be able to search doctors by specialization, location, or availability.
- Patient should be able to view doctor profiles.
- Patient should be able to book appointments based on available time slots.
- Patient should be able to view appointment status (pending / accepted / rejected / completed).
- Patient should be able to cancel appointments if needed.
- Patient should receive confirmation or rejection updates.

Non-Functional Requirements

- The interface should be easy to use and responsive.
- Login and personal data should be secure.
- Booking process should be fast and error-free.
- System should handle multiple patient requests efficiently.
- Data should be stored reliably without loss.

B. Doctor Module

Functional Requirements

- Doctor should be able to register and log in.
- Doctor should be able to create and update profile.
- Doctor should be able to set available slots.
- Doctor should be able to view appointment requests.
- Doctor should be able to accept or reject appointments.
- Doctor should be able to see patient details for booked appointments.

- Doctor should be able to manage schedule availability.

Non-Functional Requirements

- Dashboard should be simple and easy to navigate.
- Appointment updates should happen in real time or near real time.
- The system should prevent double booking.
- Doctor data and appointment details should remain private and secure.
- The module should work smoothly on desktop and mobile devices.

1.1 Input–Process–Output Table

A. Patient Registration/Login

Inputs
- Name
- Email
- Password
- Phone number
- Other profile details

Processing
- Validate entered details
- Check if email already exists
- Store patient data in database
- Authenticate credentials during login

Outputs
- Registration success message
- Login success or error message
- Redirect to patient dashboard

B. Doctor Registration/Login

Inputs
- Doctor name
- Email
- Password
- Specialization
- Experience
- Available timings

Processing
- Validate details
- Store doctor profile
- Authenticate login credentials
- Display doctor dashboard after login

Outputs
- Successful registration/login message
- Doctor profile visible in system
- Dashboard access

1.2 Test Objectives

General Test Objectives for MediConnect

- To verify that all modules work according to the specified requirements.
- To ensure that patients can successfully register, log in, search doctors, and book appointments.
- To ensure that doctors can manage availability and accept/reject appointments correctly.
- To verify that appointment scheduling works without conflicts or double booking.
- To ensure that all user data is stored and retrieved correctly from the database.
- To check that the system handles valid and invalid inputs properly.
- To verify role-based access for patients, doctors, and admin.
- To ensure the website is secure, responsive, and user-friendly.
- To confirm that appointment status updates are displayed correctly to both users.
- To identify and fix defects before deployment.

2. Test Planning

2.1 Types of Testing

1. Unit Testing

Objective:

To ensure that each small module works correctly on its own.

2. Integration Testing

Objective:

To ensure smooth communication between frontend, backend, and database.

2.2 Test Scope

1. **Authentication Module**
   - User registration
   - Login/logout
   - Role-based access
2. **Patient Module**
   - Patient profile creation
   - Doctor search
   - Appointment booking
   - Appointment viewing and cancellation
3. **Doctor Module**
   - Doctor profile management
   - Slot availability management
   - Appointment approval/rejection
4. **Appointment Management Module**
   - Appointment request generation
   - Appointment history
5. **UI/UX Testing**
   - Form behavior
   - Button functionality
   - Navigation flow
   - Responsiveness

2.3 Test Environment

Parameter	Details
OS	Windows 10 / Android
Browser	Chrome, Edge
Device	Laptop, Mobile
Backend	Node.js / Flask (as per your project)
Database	MongoDB / MySQL

2.4 Test Plan Summary

Test Type	Objective	Tools Used
Unit Testing	Test individual modules	Manual
Integration Testing	Check module interaction	Postman
System Testing	Full system validation	Browser
UAT	User validation	Manual

3. Unit Testing

3.1 Unit Test Cases

Test Case ID	Module	Input	Expected Output	Actual Output	Status
UT01	Login	Valid credentials	Login success	Login success	Pass
UT02	Login	Invalid password	Error message	Error shown	Pass
UT03	Crop	Valid input	Crop suggestion	Correct output	Pass
UT04	Crop	Empty input	Validation error	Error shown	Pass

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4. Integration Testing

4.1 Integration Test Cases

Test Case ID	Modules Involved	Scenario	Expected Result	Status
IT01	Frontend + Backend	Login request	Response received	Pass
IT02	Backend + DB	Store user data	Data saved	Pass
IT03	API + UI	Weather fetch	Data displayed	Pass

5. System Testing

5.1 System Test Cases

Test Case ID	Scenario	Expected Result	Status
ST01	Full login flow	User enters dashboard	Pass
ST02	Appointment booking	Appointment Booked	Pass
ST03	Navigation	Smooth navigation	Pass

5.2 Performance Check

Metric	Result
Page Load Time	< 2 sec
API Response Time	< 1 sec

6. User Interface and Usability Testing

6.1 UI Testing Table

Element	Test	Result
Buttons	Click response	Working
Forms	Input validation	Working
Links	Navigation	Working

6.2 Usability Observations

- Interface is simple and user-friendly
- Navigation is intuitive
- Error messages are clear

UI Home Page

7. Security and Data Validation Testing

7.1 Security Testing

Test Case	Description	Result
Login Security	Incorrect password handling	Secure
Authorization	Restricted access	Working
Data Encryption	Sensitive data	Protected

7.2 Validation Testing

Input Type	Test	Result
Empty fields	Error message	Pass
Invalid data	Rejected	Pass
SQL Injection	Prevented	Pass

8. Bug Reporting and Re-testing

8.1 Bug Report Table

Bug ID	Description	Module	Status
B01	Login crash on empty input	Auth	Fixed
B02	Incorrect time	time Module	Fixed

8.2 Re-testing

- Fixed bugs were tested again
- No new issues were found
- System stability improved

9. Test Documentation

9.1 Test Result Summary

Total Test Cases	Passed	Failed
15	15	0

9.2 Deliverables

- Test case tables
- Test result tables
- Screenshots of:
  - Successful tests
  - Failed tests
- Bug reports

9.3 Conclusion

The Mediconnect system was tested thoroughly using multiple testing methods. All modules are functioning correctly, and the application meets both functional and non-functional requirements. The system is reliable, user-friendly, and ready for deployment.